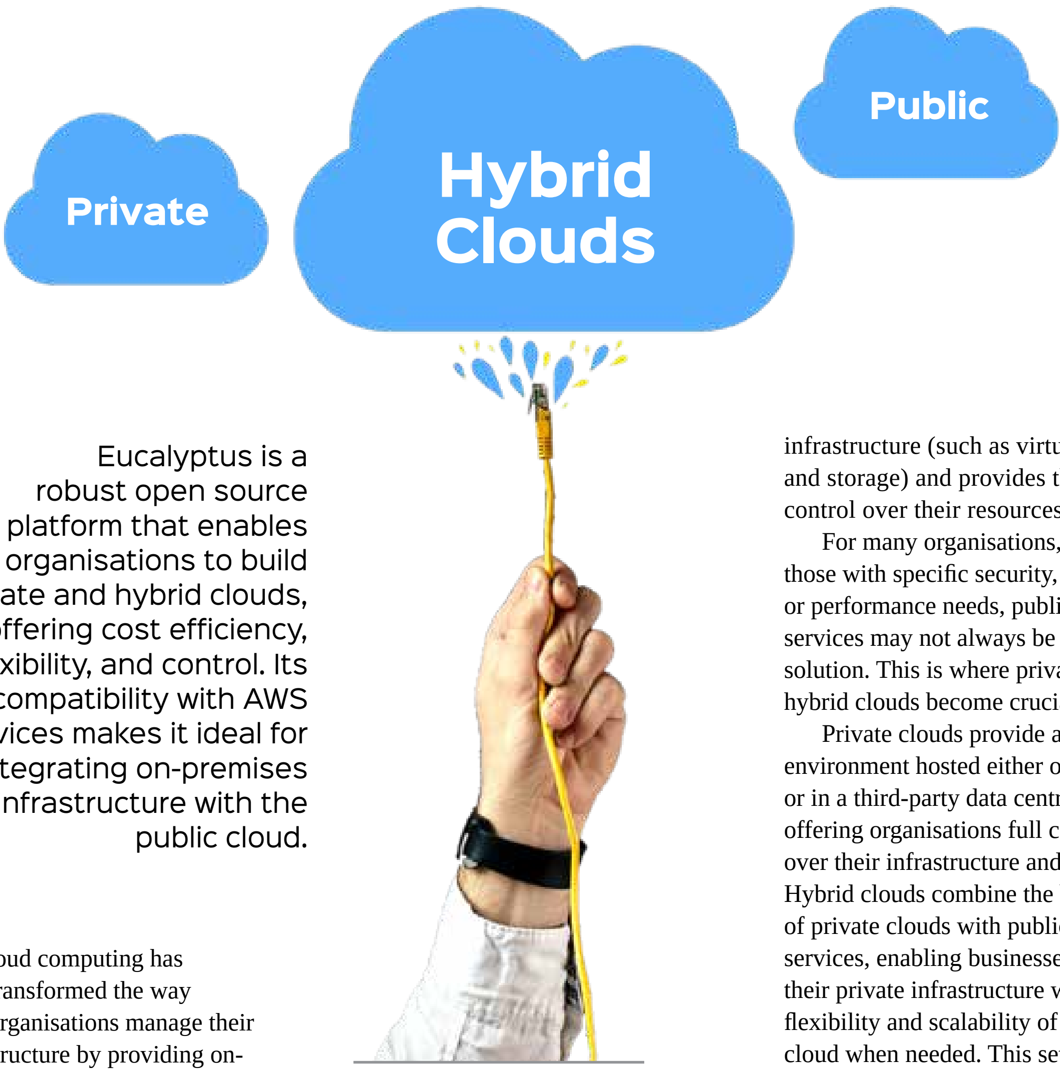
Building AWS-Compatible Private and Hybrid Clouds with Eucalyptus



Eucalyptus is a robust open source platform that enables organisations to build private and hybrid clouds, offering cost efficiency, flexibility, and control. Its compatibility with AWS services makes it ideal for integrating on-premises infrastructure with the public cloud.

Cloud computing has transformed the way organisations manage their IT infrastructure by providing on-demand access to computing resources such as servers, storage, databases, and software over the internet. This eliminates the need for physical hardware and allows businesses to scale resources according to their needs, which in turn reduces operational costs, improves efficiency, and enhances flexibility.

Cloud computing operates on three primary service models: infrastructure as a service (IaaS), platform as a service (PaaS), and software as a service (SaaS). These models offer varying levels of control and flexibility for businesses. IaaS, in particular, allows organisations to rent IT infrastructure (such as virtual machines and storage) and provides the most control over their resources.

For many organisations, especially those with specific security, compliance, or performance needs, public cloud services may not always be the ideal solution. This is where private and hybrid clouds become crucial.

Private clouds provide a dedicated environment hosted either on-premises or in a third-party data centre, offering organisations full control over their infrastructure and data. Hybrid clouds combine the benefits of private clouds with public cloud services, enabling businesses to extend their private infrastructure with the flexibility and scalability of the public cloud when needed. This setup helps manage workloads dynamically, optimise costs, and achieve greater operational efficiency.

Eucalyptus is an open source software platform that enables businesses to build private and hybrid cloud environments. It provides organisations with a cost-effective solution for implementing cloud